University of East London

FCIS – Ain Shams University
Subject: AIT 417 Introduction to Vision and Robotics
Exam: (Final) 15/01/2022
Year: (1st term) 4th undergraduate

Examiner: Dr. Dina Khattab
Program: Artificial Intelligence
Academic year: 2021-2022
Duration: 2 hours

Answer the following four questions: (Total marks: 50)

1st Question marks: 17

A) **[12 marks] State whether each of the following statements is True or False. Correct False statements.**

1. If you are unsure of how many clusters you have in your data, the best method to do clustering would be K-means.
2. In a Seimese network, the definition of $\max(\|f(A)-f(P)\|_2^{\,2} - \|f(A)-f(N)\|_2^{\,2} + \alpha,\ 0)$ would be a correct definition for a triplet loss (where A represents the anchor image, P represents the positive image and N represents the negative image), considering that $\alpha \geq 0$.
3. Autoencoders can be used for dimensionality reduction.
4. The Region Proposal Network (RPN) is the process of region proposals used as part of the Fast-RCNN object detection algorithm.
5. While applying downsampling of an image in the hierarchical LK algorithm, the processes of image sharpening would help avoid aliasing.
6. The SIFT feature descriptor is the tool used for extracting image components that are useful for the representation and description of the object shape in an image.
7. The brightness constancy constraint that is utilized in optical flow computation relates the flow to both the spatial and temporal gradients of the image sequence.
8. In an object detection model where it is supposed to produce a bounding box for each object that appears in the image, you consider a detection output as a false positive if IoU < 0.5.
9. Normalizing the SIFT descriptors in the direction of the dominant gradient orientation would allow the descriptors to be invariant to image translation.
10. The process of instance segmentation is the process which assigns a different label to the pixels of each object instance in an image.
11. Autoencoders is a recommended model for feature representation in labeled data.
12. In Content-Based Image Retrieval (CBIR), the retrieval upon features extracted from the former layers of CNN would generate high mean Average Precision (mAP) result for the task of semantic similarity image retrieval.

B) **[5 marks]** Consider estimating the optical flow given two images I(x, t) and I(x, t+1). The brightness constancy constraint provides a linear equation for a given image pixel in the two consecutive images of an image sequence.

i. **What** is the brightness constancy constraint? **What** are its assumptions?
ii. **Explain** an algorithm for the optical flow estimation between the two images. **State** any extra assumptions you may need to let this algorithm work. **Show** the used equations in matrix form.
iii. Given that the motion between the two images is non-linear, **what** are the approximated linearized version of your optical flow estimation algorithm that can be used to update the motion estimation? **State** the algorithm steps.

2nd Question — marks: 13

A) **[7.5 marks]** Consider 12 points are painted on a dark plane. 5 points are colored pure red, 5 are colored pure green, and 2 are colored white. Two cameras (stationed with a translational shift) take pictures of the plane. One camera is equipped with a red filter (doesn't see green color) and one camera is equipped with a green filter (doesn't see red color).

i. **Which** points does each camera see?

ii. Do you think that enough point correspondences are available to recover a transformation (homography) between the two images? **Explain** why.

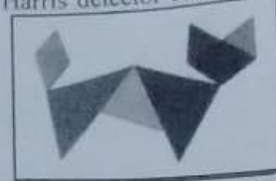
iii. **Explain** an algorithm that can compute the set of good point correspondences between the two images. Define the used parameters of the algorithm, and how the value of each parameter should be selected.

B) **[5.5 marks]** Assume you have an image containing different geometric shapes with different sizes and orientations. Your task is to locate the corners of these objects. Describe the Harris detector for corner detection by giving:

i. The error equation.

ii. The second-moment matrix and how it is used to select good corners.

iii. The main steps of the Harris algorithm



3rd Question — marks: 10

A) **[6 marks]** Assume that you have been given images of the road taken from a camera mounted on the front of a car. Your task is to find the road using semantic segmentation.
Design a network based on CNN to implement this task. **Explain** and **Sketch** a general architecture diagram in terms of input, core and output components.

B) **[4 marks]** **Compare** between Single Object Tracking (SOT) and Multiple Object Tracking (MOT) in terms of:

- Short/long term tracking.
- Detection-free/Detection-based tracking.
- Type of tracking errors.
- Evaluations Metrics.

4th Question — marks: 10

A) **[5 marks]** Your job is to build a computer vision system to track a news reporter's face. Given any video clip (which includes one person in it who is facing the camera but you don't initially know exactly where), your application should implement a ***Face localization algorithm*** based on **skin color**. The output of the localization algorithm should be a bounding box containing the news reporter's face.
Once the face is localized in each video frame, a ***Tracking algorithm*** should output a trajectory of the face in the entire video (sequence of x, y locations for the center of the face) per video clip.
Describe a complete approach to accomplish this task, including:

i. Description of the ***Face localization algorithm*** in terms of:
- Proposed training dataset.
- Preprocessing of data.
- Training.
- Testing on the given video.

ii. Description of the ***Tracking algorithm*** per video clip.

B) **[5 marks]** One important application of autoencoders is image denoising, where the input images are corrupted by adding some noise and feed them to the autoencoder to learn to remove the unwanted information. **Sketch** a general architecture of the denoising autoencoder showing the construction of the different layers of Encoder, Code and Decoder parts. **Explain** the role of each layer.

P(skin) P(not skin)

With My Best Wishes Dr. Dina Khattab

~~P(skin)~~

$P(x \mid skin) = \frac{P(x \cap skin)}{P(skin)}$

$P(skin \mid x) = \frac{P(x \mid skin)P(}{P(x)}$